

Caleb Elder

706-612-7923 | caleb.elder4@icloud.com | in/calebelder-compsci | github.com/ucgmae47

Education

University of Georgia

May 2027

4.0 GPA | B.S. Computer Science and B.S. Applied Mathematics

Relevant Coursework (Completed): Software Development, Data Structures, Theory of Computing, Systems Programming, Discrete Mathematics, Applied Linear Algebra, Mathematical Analysis

Relevant Coursework (In Progress): Software Engineering, Algorithms, Computer Architecture, Computer Networks, Graph Theory

Technical Skills

Programming Languages: Proficient in Java, Python, C | Skills in SQL, C++, JavaScript, HTML, CSS, LaTeX

Languages: Proficient: Spanish | Intermediate: French, Chinese | Conversational: German, Russian, Korean

Developer Tools: Git, GitHub, Docker, VS Code, IntelliJ, PyCharm, Jupyter Notebook, Maven, Linux, Bash, Zsh

AI Tools: Cursor, OpenAI Codex, GitHub Copilot, Claude

Libraries: React, JavaFX, Gson, NumPy, Pandas, PyTorch, Flask, SQLAlchemy, Jinja2

Technologies: REST API, JSON, MySQL

Experience

AI4STEM (VIPR Program)

August 2025 – December 2025

Undergraduate Research

Athens, GA

- Researched LLM-based automated assessment and sourced datasets from GitHub, HuggingFace, and Kaggle to identify gaps and define project goals for improving student feedback
- Designed a multimodal rubric integrating text and image data for holistic student assessment
- Developed Python scripts in PyCharm and Jupyter Notebook to process student Excel data into JSON and leverage the GPT-5 API to produce automated, reasoning-based scores
- Calculated inter-rater reliability using Cohen's Kappa and compared LLM scores with teacher scores
- Collaborated on code development and version control using GitHub and prepared a CURO Symposium presentation detailing the development process (software design, architecture, data pipeline) and research findings

Rachel's Southern Style Restaurant

June 2021 – July 2025

Server

Watkinsville, GA

- Promoted through 3 positions (busser → cashier → server), reflecting strong work ethic and reliability
- Managed workload of up to 40 hours/week, demonstrating strong time management and dedication
- Trained and supervised 15–20 team members on customer service, workflow efficiency, and closing procedures

Projects

Chinese Language Learning App | *Git, Java, JavaFX, Cohere API, Dictionary API, JSON, Gson* April 2025

- Developed a JavaFX app generating AI-powered Chinese reading practice stories tailored to HSK proficiency
- Integrated Cohere API for generating leveled text and Chinese Character API for hover-over translations
- Parsed JSON data using Google's Gson library, enabling efficient API data extraction and story generation
- Applied inheritance, polymorphism, and generics to build a modular, object-oriented design

Athens Figure Skating Club Website | *Python, Flask, MySQL, HTML, CSS, JavaScript* Ongoing / 2026

- Built a full-stack platform for club members to view events, profiles, and schedules interactively
- Developed a secure user authentication system with password hashing and role-based access to protect data
- Designed front-end templates with Jinja2 and CSS, integrating dynamic content from the back-end
- Implemented ORM with SQLAlchemy for database interactions, including schema management and migrations
- Managed database integration using MySQL, handling event scheduling, user profiles, and admin content
- Connected front-end UI with back-end Flask routes to deliver a cohesive, full-stack web experience
- Deployed on Render with MySQL database on Aiven and Backblaze B2 cloud storage for media and static assets

Extracurriculars

Ultimate Frisbee Summer League | Teamwork, Strategy, Communication

June 2025 – July 2025

- Built strong connections through teamwork, communication, and sportsmanship
- Learned and applied strategic gameplay techniques to improve performance and team coordination